Samarvir Garg

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Education

University of Toronto Toronto, ON
Bachelor of Applied Science and Engineering – Computer Engineering *Sep 2025 – Jun 2030*
- Relevant Coursework: Data Structures and Algorithms, Computer Organization, Digital Systems, Differential Equations, Linear Algebra

Technical Skills

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS
Frameworks: Spring Boot, TensorFlow, Flutter, Firebase, REST APIs
Developer Tools: Git, AWS, Power BI, VS Code, IntelliJ, Arduino
Libraries: OpenAI API, NumPy, Pandas

Experience

AI Engineer Trainee May 2026 – Present
Trakop – Retail Supply and Logistics Platform *Toronto, ON*
- Developed AI powered solutions for logistics and business automation using Python, TensorFlow, and OpenAI APIs, improving delivery route efficiency and inventory accuracy
- Built and tested predictive models for inventory forecasting, delivery route optimization, and payment risk analysis on a live SaaS platform
- Collaborated with cross functional teams using Git for version control and AWS for deployment, following agile development practices

Member, Driverless Deep Perception Subteam Jan 2026 – Present
University of Toronto Formula Racing *Toronto, ON*
- Developing ML based perception pipeline for cone detection, classification, and localization using camera image streams in a real time autonomous racing environment
- Collecting and manually labeling training data to improve model generalization across varied track conditions
- Evaluating model performance using IoU, mAP, and inference latency metrics on NVIDIA Jetson hardware under real time constraints

High School Research Intern Nov 2024 – Feb 2025
University of Toronto, Dept. of Psychiatry *Toronto, ON*
- Selected to conduct research under Prof. Clement C. Zai on pharmacogenetic reporting and data visualization in mental healthcare as a high school student
- Developed AI powered clinical dashboards and decision support tools using large language models to automate pharmacogenetic data reporting for clinicians
- Authored a technical report proposing scalable, data driven visualization design recommendations for clinical use

Projects

Emergency Route Optimization System | *Java, Spring Boot, SQLite, REST API* Jan 2026
- Designed and built a full stack backend system using Java Spring Boot with a 3 layer (Controller, Service, Repository) architecture to handle emergency dispatch routing
- Developed a REST API with CRUD operations to manage emergency inputs, ambulance updates, and routing decisions in real time
- Integrated SQLite for persistent storage of emergency data and simulation logs, and validated the system through a Becker Robots simulation frontend; placed 3rd overall at UTEK 2026

Autonomous Robotics System | *C++, Arduino Uno R4 Minima* Jan 2026
- Built an autonomous Arduino robot for a competition course involving navigation, object pickup, and target alignment in a structured environment
- Integrated IR sensors, ultrasonic ranging, and color sensing for line following, obstacle detection, and path based behavior selection
- Implemented a modular C++ architecture separating hardware control, sensor perception, and high level behaviors to improve reliability and debugging speed
- Placed 2nd overall and 1st for top track performance out of 60 plus teams at UTRA HACKS 2026

UniQuery | *Flutter, Firebase, Cloud Firestore, Firebase Auth* Jun 2026 – Present
- Designed and built a cross platform academic query management system for students, professors, TAs, and admins using Flutter and Firebase
- Implemented role based access control with Firebase Authentication and Firestore security rules, giving each user role a distinct dashboard and permissions
- Built file upload and notification features using Firebase Storage and Firebase Cloud Messaging for real time communication
- Tested and debugged the application on a physical Android device, handling edge cases for offline access and varied screen sizes